

Catching Cross-Generator Email Impersonation with Content-Invariant Authorship Embeddings

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Abstract

Business email compromise increasingly relies on large language models (LLMs) to imitate a specific person’s writing style. We describe a sender-verification system that enrolls each sender’s style as a centroid in an authorship embedding space and flags messages whose style does not match the claimed sender. The central technical difficulty is *cross-generator generalization*: a detector trained against forgeries from one LLM learns generator-specific artifacts and collapses on forgeries from unseen models (TPR@1% $0.91 \rightarrow 0.045$). We show that the two obvious drop-in fixes both fail on this threat — zero-shot machine-generated-text detectors score at or below chance (AUC 0.49–0.54) on few-shot *imitation* emails, and an off-the-shelf content-independent style embedding underperforms our fine-tuned encoder on every axis. The fix that works is retraining the authorship backbone itself, with two levers: (i) synthetic hard negatives from *multiple* generator vendors, evaluated only on *held-out* vendors, and (ii) identity diversity, expanding training authors from 44 to 844. A completed 2×2 factorial shows the levers are *super-additive*: identity diversity alone does nothing for forgery detection, but it amplifies the synthetic-negative gain from +0.23 to +0.37 AUC. The final model reaches AUC 0.980 ± 0.003 on forgeries from generators never seen in training (91% caught at a 5% false-alarm budget), holds a wrong-human false-accept guardrail at 8%, transfers to novel vendors outside both training and evaluation (AUC 0.996), and improves cross-topic authorship verification (PAN $0.779 \rightarrow 0.857$). All results are reported as mean $\pm$ std over evaluation seeds, with a training-seed reproduction.

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1 Introduction

Impersonation is the core mechanic of business email compromise (BEC): a message arrives that claims to be from a colleague or executive, and its content and address look legitimate. Modern LLMs sharply lower the cost of the strongest form of this attack — given a handful of a victim’s real emails, an LLM can produce fluent text *in that person’s approximate style*, defeating both content-based filters and human suspicion.

Our system defends at the layer the attacker must ultimately forge: **authorship**. Every sender in an organization is enrolled by encoding their verified historical emails into a style embedding space and storing a per-sender profile (centroid, spread, and count). An incoming message claiming to be from that sender is encoded and scored by its distance to the profile; large deviations are flagged. The encoder is trained once, on public corpora — it never sees the deployment organization’s senders at training time and is never retrained at deployment; new senders are enrolled with forward passes only.

This paper documents the system and, more importantly, the empirical path to making it work against LLM forgeries. Three findings organize the paper:

1. **The central problem is cross-generator generalization (§6)**. Training against forgeries from a single LLM produces a detector of *that LLM’s artifacts*, not of forgery: on

forgeries from unseen generators, the catch rate at 1% false-alarm collapsed from 0.91 to 0.045. Consequently, every headline number in this paper is computed on forgeries from generators *excluded from training* (Claude 3.5 Haiku and Gemini 2.5 Flash), and we additionally spot-check vendors outside both training and evaluation.

2. **Drop-in alternatives fail on imitation (§6.2).** Zero-shot machine-generated-text (MGT) detectors (Fast-DetectGPT, Binoculars) score at or below chance on few-shot imitation emails — text prompted to mimic a specific human is precisely the case the MGT literature warns about. An off-the-shelf content-independent style embedding (StyleDistance) is worse than our fine-tuned encoder on every evaluation slice. The fix has to be trained into our own authorship backbone.
3. **Two levers close the gap, and they interact super-additively (§7–9).** Multi-vendor synthetic hard negatives are the primary lever (+0.23 AUC); expanding training identities from 44 to 844 authors does nothing *by itself* for forgery detection but amplifies the synthetic lever to +0.37, yielding held-out AUC 0.980 ± 0.003 . Identity diversity’s standalone payoff is content-invariance — a separate axis on which it improves cross-topic verification from 0.779 to 0.857–0.879.

We also report what did not work and what remains open: a scorer that Goodharts its threshold when synthetics become easy (§8.2), a length-invariance weakness that survives every version (§10), and the evaluation practices — held-out generators, a wrong-human guardrail metric, multi-seed error bars, a training-seed reproduction — that we believe are the minimum for trusting results in this adversarial setting.

2 Threat model and task

Setting. A deployment organization provides a historical archive of verified-genuine email. The system runs in two phases. *Enrollment*: each sender’s emails are preprocessed (reply chains and signatures stripped; URLs, addresses, and phone numbers masked), encoded, and summarized into a profile $\{\mu_s, \sigma_s, k_s\}$ — centroid, mean cosine spread, and email count. *Scoring*: an incoming email claiming sender s is encoded and scored against $\{\mu_s, \sigma_s, k_s\}$, returning a score in $[0, 1]$ and a verdict (genuine / suspicious / abstain when $k_s < 5$).

Adversaries. Two impostor classes claim sender s ’s identity:

- **Wrong-human impostors**: mail genuinely written by a different person. This is the classical authorship-verification negative and is comparatively easy for our encoder (AUC ≈ 0.97 across all versions).
- **LLM forgeries**: text produced by an LLM *prompted with several of s ’s real emails* as style exemplars (few-shot imitation). This models an attacker who has compromised or scraped samples of the victim’s writing — the realistic BEC case — and is the hard class.

We assume the attacker can use any current or future LLM vendor, so the defender cannot count on having trained against the attacker’s generator. We do not assume the attacker has query access to the deployed detector (adaptive attacks are out of scope; see §10).

Operating requirements. The deployment metric is the catch rate at a small false-alarm budget, TPR@5% (secondarily TPR@1%), on forgeries from *held-out* generators, subject to a **guardrail**: at the chosen threshold, the false-accept rate on wrong-human impostors, FPR_{other} , must stay ≤ 0.10 . A system that catches forgeries by flagging everything unusual — including legitimate mail from a colleague — is not deployable.

3 Related work

Authorship representation. LUAR (Rivera-Soto et al., 2021) learns universal authorship representations by contrasting episodes (bags of texts) by the same author, pretrained at Reddit scale; it is the backbone we fine-tune. Content-independent style embeddings (Wegmann et al., 2022; Patel et al., 2024) attack the confound that same-author text pairs also share topic; StyleDistance trains on synthetic paraphrase pairs that hold content fixed while varying style. Cross-topic authorship verification is the standard stress test (Kestemont et al., 2020).

Machine-generated-text detection. Zero-shot detectors score text by likelihood statistics under reference models — Fast-DetectGPT’s conditional probability curvature (Bao et al., 2023) and Binoculars’ cross-perplexity ratio (Hans et al., 2024). Benchmarks consistently find that *trained* MGT detectors fail to transfer across generators (Dugan et al., 2024), and that adversarial or stylized generations defeat zero-shot detectors. Our results sharpen this for the impersonation threat: few-shot imitations of a specific human are *more* human-like than real email under these detectors’ statistics (§6.2).

Positioning. Our approach is personalized authorship verification used as a forgery detector: rather than asking “was this text machine-generated?” (generator-dependent, brittle), we ask “was this text written by the person it claims to be from?” — a question whose signal belongs to the *sender*, not to the attacker’s generator. The project’s arc is evidence that this is the durable formulation.

4 System

4.1 Encoder

The backbone is LUAR-MUD (Rivera-Soto et al., 2021) with episode pooling: K emails from one sender form an episode and produce one L2-normalized style embedding (projection dim 128, max length 512 tokens). We fine-tune with LoRA (Hu et al., 2022) ($r=16$, $\alpha=32$, dropout 0.05) on the query/key/value projections; the backbone stays frozen. Training batches are built by a $P \times K$ sampler (P senders $\times K=8$ emails, batch size 128) with random crop augmentation (30% of texts cropped to 5–60 words) to reduce length sensitivity.

Why an authorship backbone. In an early bake-off, topic-pretrained encoders (RoBERTa, MPNet) trained with the same contrastive recipe reached validation AUROC ≈ 0.53 — near chance — while LUAR-MUD reached 0.956. Masked-LM embeddings cluster by topic, and no amount of LoRA fine-tuning on 44 senders fixed the inductive bias; authorship pretraining is load-bearing. (A diagnostic: RoBERTa’s same-sender similarity was high, but so was its different-sender similarity — nothing discriminative was learned.)

4.2 Objective

The loss is an episodic supervised-contrastive objective: within each batch, embeddings sharing a sender label attract and all others repel, with temperature $\tau=0.05$ (Khosla et al., 2020). Episodes are formed with variable support size (2–6 emails), and a plain SupCon term is mixed in with weight 0.5. Lower temperature weights the hardest negatives most — which, by construction (below), are LLM imitations of the anchor sender.

4.3 Synthetic hard negatives

For each training sender we generate *imitation* emails: an external LLM is prompted with several of the sender’s real emails and asked to write new mail as that person. Each imitation set is registered as its own identity, `alice__syn`, distinct from `alice`. A `SyntheticBalancedSampler` guarantees $n_{\text{syn}}=4$ (real, synthetic) sender pairs per batch, so every batch forces the loss to separate real-Alice from LLM-Alice — the model must find stylistic features the LLM failed to reproduce.

Two rules were learned the hard way and are enforced in code:

- **LLM text is never a positive.** An earlier version used LLM-generated cross-register text as additional positives for the sender; it regressed wrong-human verification and was banned (confirmed by a five-arm ablation isolating the objective change).
- **Checkpoint selection must not watch synthetics alone.** A synthetic-only model monitor selects checkpoints that leak up to 60% of wrong-human mail. The training monitor is $\min(\text{pAUC}_{\text{other}}, \text{pAUC}_{\text{synthetic}})$ at 5% FPR, the worst case of the two impostor axes.

4.4 Scoring

At inference, the query embedding q is compared to the claimed sender’s profile. Two scorers matter in this paper:

- **linear_z3** (baseline): $z = (1 - \cos(q, \mu_s))/\sigma_s$, $\text{score} = \max(0, 1 - z/3)$. Simple, no per-sender fitting beyond spread.
- **mahalanobis** (deployed): per-sender Mahalanobis distance under a Ledoit–Wolf shrunk covariance (Ledoit and Wolf, 2004) fit to the enrollment embeddings, gated by enrollment count: senders with $k < 5$ fall back to cosine ranking (a covariance cannot be estimated), and $k < 5$ also triggers the abstain tier.

Both scorers *rank* wrong-human impostors nearly identically (AUC 0.94–0.97); they differ in where a synthetic-anchored threshold lands, which decides the guardrail (§8.2).

5 Evaluation protocol

The held-out-generator rule. Forgeries used in training come from **GPT-4o-mini** and **Llama-3.1-70B** (plus DeepSeek in one version). Forgeries used for *all headline evaluations* come from **Claude-3.5-Haiku** and **Gemini-2.5-Flash**, which are never used in training, for any version. A further spot-check uses **Qwen-2.5-72B** and **DeepSeek-V3** — outside both the training *and* evaluation sets (§9.3). Every version of the system is graded on the same held-out pool.

Probe. The evaluation pool profiles 44 Enron senders (the post-filter maximum) with $k=8$ enrollment emails each, and scores 264 genuine queries, 327 held-out LLM forgeries (172 Claude, 155 Gemini), and 600 wrong-human queries. Metrics: AUC (forgery vs. genuine), TPR@5% and TPR@1% at synthetic-anchored thresholds, $\text{FPR}_{\text{other}}$ at the 5% threshold (the guardrail), and AUC on genuine-vs-wrong-human ranking. Each metric carries a 1000-sample paired bootstrap; in addition, the entire probe (sender, enrollment, query, and impostor draws) is repeated over **5 seeds** and reported as $\text{mean} \pm \text{std}$ throughout.

The in-distribution trap. Per-run validation panels (negatives drawn from that run’s *own* training generators) are useful for watching convergence and useless for comparison: the single-generator model posts validation $\text{TPR@1\%} = 0.74$ while catching 4.5% of held-out forgeries. All numbers in this paper come from the fixed held-out probe above, never from training dashboards.

Secondary axes. Content-invariance is measured on PAN 2020 cross-topic authorship verification pairs (Kestemont et al., 2020) (clean out-of-domain: never trained on) and a held-out 150-author slice of the Blog Authorship Corpus (Schler et al., 2006); length and register robustness on stratified Enron pair slices.

6 The central problem: cross-generator generalization

6.1 Single-generator training learns the generator, not forgery

The v11 model — the strongest single-generator configuration, trained with Mistral-7B-family imitations — looked excellent in distribution: $\text{TPR@1\%} = 0.91$ against Mistral forgeries. On the held-out Claude+Gemini pool the same model collapses to $\text{TPR@1\%} = 0.045$ and AUC 0.75 (multi-seed: $\text{AUC } 0.690 \pm 0.020$). The learned “LLM-ness” feature was a Mistral artifact detector. This single number reframed the project: the metric that matters is **TPR at low FPR on held-out generators**, and everything after v11 is an attempt to move it.

6.2 Failed alternative 1: zero-shot MGT detectors

If forgery detection cannot be trained per-generator, generator-agnostic zero-shot detectors are the natural drop-in. We evaluated two, at their recommended settings, on exactly our held-out pool (genuine = real Enron mail, forgeries = few-shot imitations by Claude/Gemini):

Detector	Claude	Gemini	Pooled	Len-matched	TPR@1%
Fast-DetectGPT	0.399	0.693	0.538	0.536	0.012
Binoculars	0.398	0.582	0.485	0.523	0.016
<i>Authorship signal (v12, ref.)</i>	<i>0.829</i>	<i>0.746</i>	<i>0.854</i>	—	<i>0.13</i>

Table 1: Zero-shot MGT detectors on few-shot imitation emails (held-out pool); all columns except TPR@1% are AUC. Both put Claude imitations *below chance*: the imitations are more “human-like” under likelihood statistics than real, messy corporate email. Length-matching does not rescue them.

This is not a weak-detector artifact — two unrelated methods agree, after length control. Few-shot imitation produces fluent, low-perplexity text *by construction*; “is this generic LLM text?” is the wrong question for this threat. It also reinterprets what v11–v13’s trained detector

head was doing: not detecting LLM-ness, but generator artifacts — explaining both its in-distribution strength and its held-out collapse.

6.3 Failed alternative 2: off-the-shelf style embeddings

If the issue is content sensitivity, a content-independent style embedding trained elsewhere might substitute for ours. StyleDistance (Patel et al., 2024), evaluated over the same out-of-domain pair files as our encoder, is worse on every slice — including the cross-topic slice its training specifically targets:

Slice (pairwise AUC)	StyleDistance	Fine-tuned LUAR (v12)	Δ
PAN cross-topic	0.748	0.779	−0.031
Blog	0.793	0.839	−0.046
Mixed-length pairs	0.580	0.931	−0.351
Claude / Gemini forgery	0.560 / 0.583	0.829 / 0.746	−0.27 / −0.16

Table 2: An off-the-shelf content-independent embedding does not transfer to email. The lesson is not that content-independence is wrong — it is that it must be trained into *our* encoder, not imported.

Together, §6.2 narrows the design space to one path: **retrain the authorship backbone itself** to be generator-robust and content-invariant. These two negative results were obtained *before* committing GPU budget to that path, and they are why the rest of the paper is about training levers rather than architecture swaps.

7 Method: two levers

Each version changes one thing; all are graded on the identical held-out probe (Table 3).

v12 — generator diversity. Train the synthetic hard negatives with *two* vendors (GPT-4o-mini + Llama-3.1-70B), holding out Claude+Gemini. Hypothesis: diversity in the training adversary transfers to unseen adversaries. Result: AUC 0.690 $\rightarrow$ 0.871; per-generator Claude 0.639 $\rightarrow$ 0.829, Gemini 0.614 $\rightarrow$ 0.746. Confirmed — the first real fix.

v13 — more generator diversity (plateau). Add DeepSeek as a third vendor with more per-sender volume. Result: 0.872 ± 0.021 — statistically flat versus v12. Generator count had stopped paying; the bottleneck was elsewhere. The diagnosis: with only 44 training identities, the model’s notion of “this person’s style” is too coarse to distinguish a good imitation from the real author, regardless of how many generators supply imitations.

v14 — identity diversity (a clean split of effects). Expand training identities from 44 Enron senders to **844 authors** by adding 800 authors from the Blog Authorship Corpus, forcing the contrastive objective across far more authors and topics. This run dropped synthetics (a sampler limitation, fixed in v14b). Result: one breakthrough and one collapse. Content-invariance moved for the first time in the project — PAN cross-topic $0.779 \rightarrow 0.879$ — while forgery detection collapsed to 0.610 ± 0.016 (Claude $0.829 \rightarrow 0.541$), proving synthetics are load-bearing for forgery-catching. As an accidental ablation, v14 cleanly separates the levers: *identity diversity buys content-invariance; synthetics buy forgery detection*.

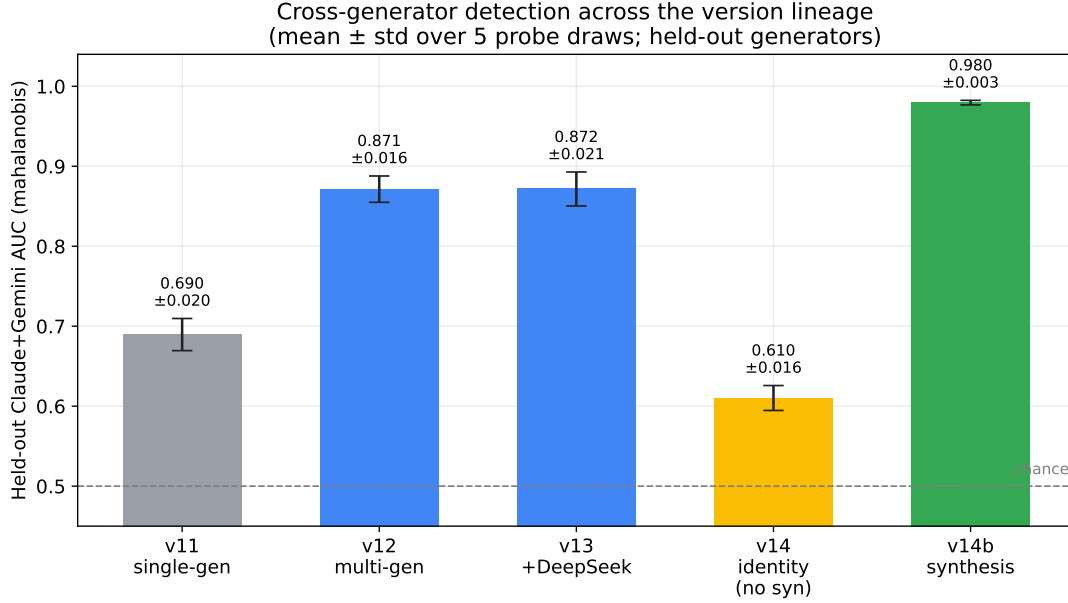


Figure 1: Held-out Claude+Gemini forgery AUC (Mahalanobis scorer), mean±std over 5 probe seeds. Every version-to-version gap is far larger than its seed variance.

Version	Change	AUC	TPR@5%	TPR@1%	FPR _{other} @5	AUC g/other
v11	single generator	0.690 ± 0.020	0.121 ± 0.017	0.043 ± 0.012	0.000 ± 0.000	0.967 ± 0.003
v12	multi-generator	0.871 ± 0.016	0.453 ± 0.064	0.155 ± 0.036	0.001 ± 0.001	0.965 ± 0.002
v13	+3rd vendor	0.872 ± 0.021	0.499 ± 0.069	0.255 ± 0.047	0.001 ± 0.001	0.970 ± 0.004
v14	844 authors, −syn	0.610 ± 0.016	0.137 ± 0.015	0.060 ± 0.013	0.010 ± 0.003	0.667 ± 0.015
v14b	844 authors, +syn	0.980 ± 0.003	0.909 ± 0.017	0.533 ± 0.055	0.079 ± 0.014	0.967 ± 0.003

Table 3: Main result: held-out Claude+Gemini pool, Mahalanobis scorer, mean±std over 5 probe seeds. AUC is forgery vs. genuine; FPR_{other}@5 is the wrong-human false-accept rate at the 5% operating point (guardrail ≤ 0.10); AUC g/other is genuine vs. wrong-human ranking.

v14b — the synthesis. The blocker was the sampler: the old epoch length was set by the 44 synthetic pairs (~11 batches), starving the 800 blog authors. The fixed synthetic-balanced sampler scales the epoch to the real-author pool (~105 batches) while still guaranteeing synthetic pairs in every batch (verified byte-identical on all prior configs, which have no unpaired authors). v14b = 844 authors + GPT/Llama synthetics + the worst-case-anchored monitor. Result: **AUC** 0.980±0.003, better than every predecessor on every axis simultaneously (Table 3, Figure 1); per-generator Claude 0.972, Gemini 0.953.

8 Main results

8.1 At the operating point

At the same 5% false-alarm threshold on the held-out pool (seed-0 probe), v12 and v14b allow the same genuine mail (251/264) — but v12 misses 61% of forgeries (198/327) while v14b misses 12% (40/327), and both keep wrong-human catch rates high (598 vs. 550 of 600). The change

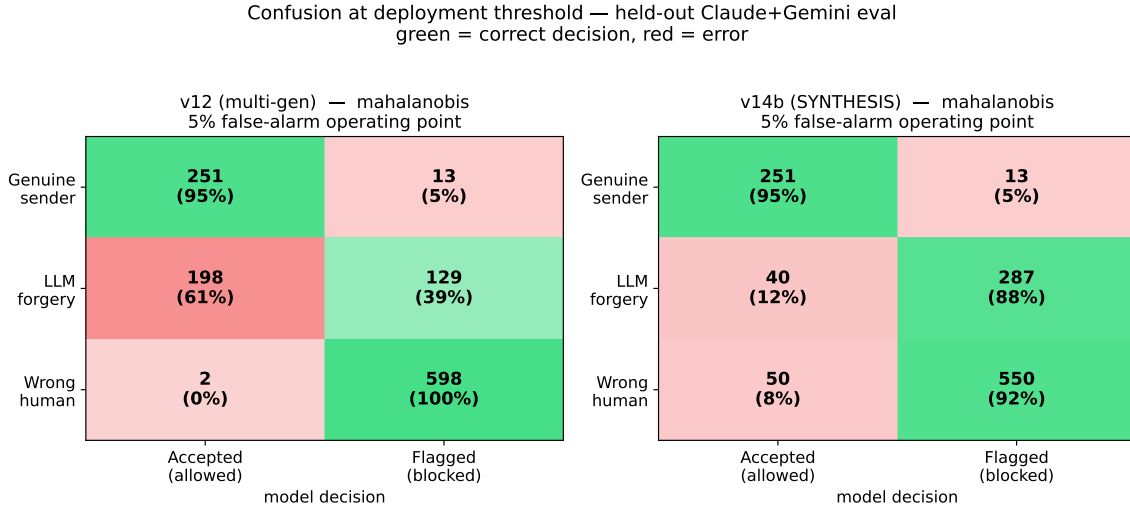


Figure 2: Confusion at the 5% false-alarm operating point, held-out pool (probe seed 0): v12 (left) vs. v14b (right).

a deployment sees is a 39% → 88% forgery catch rate at unchanged false-alarm cost, against generators the model has never trained on (Figure 2).

8.2 The guardrail: scorer choice decides deployability

v14b separates synthetics so well that a threshold anchored on the synthetic pool alone can land badly for the *wrong-human* axis. Across all 5 seeds:

Scorer	TPR@5% (forgeries)	FPR _{other} @5 (wrong-human leak)	
linear_z3	0.943 ± 0.013	0.253 ± 0.032	breaches guardrail
mahalanobis (deployed)	0.909 ± 0.017	0.079 ± 0.014	holds ≤ 0.10

Table 4: Scorer choice at the synthetic-anchored 5% threshold, mean±std over 5 seeds. Both scorers *rank* wrong-humans well (AUC 0.94–0.97); the linear scorer simply sits at a bad threshold. This is an operating-point property, invisible to AUC.

The deployment rule that follows — and that the novel-vendor result (§9.3) makes even more pointed — is: **ship the Mahalanobis scorer and anchor the threshold on the worst case of the wrong-human and synthetic axes, never on the synthetic pool alone.** The better the model gets on synthetics, the more degenerate a synthetic-only threshold becomes.

9 Ablations and robustness

9.1 Identity × synthetics: a super-additive interaction

The lineage confounds the two levers (v12 changed generators; v14 changed identities *and* dropped synthetics). We completed the factorial by training the missing corner — 44 authors without synthetics, at a step budget between v12’s and v14’s to avoid an undertraining confound:

Three readings, none visible from the lineage alone:

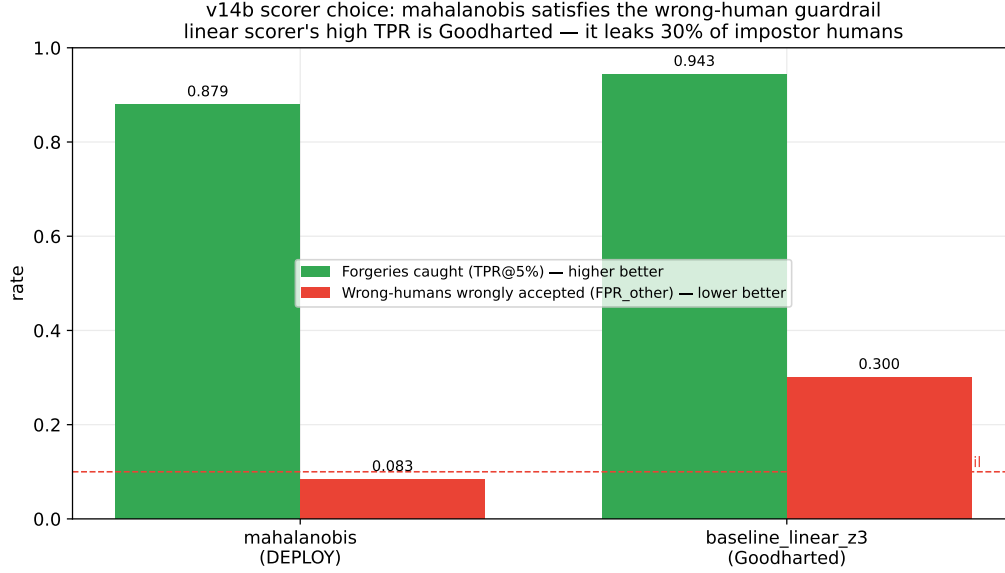


Figure 3: Scorer choice at the deployment operating point (seed-0 probe; Table 4 gives the multi-seed bands). The linear scorer catches slightly more forgeries but leaks 30% of wrong-human impostors; Mahalanobis holds the guardrail.

Held-out AUC	no synthetics	+ synthetics
44 authors (Enron)	0.640 ± 0.012	0.871 ± 0.016 (v12)
844 authors (Enron+Blog)	0.610 ± 0.016 (v14)	0.980 ± 0.003 (v14b)

Table 5: Identity × synthetic 2×2 (held-out Claude+Gemini, Mahalanobis, 5-seed mean±std).

- **Identity diversity alone does nothing for forgery detection:** 0.640 → 0.610 going 44 → 844 authors without synthetics — flat to slightly negative.
- **Synthetics are the primary lever:** +0.231 AUC at 44 authors.
- **Identity amplifies synthetics:** the synthetic gain grows to +0.370 at 844 authors. Equivalently, identity expansion only pays off in the presence of synthetics (0.871 → 0.980).

The precise claim is therefore not “two independent levers” but: *identity diversity’s value for catching LLM imitations is unlocked by synthetic hard negatives*. A sharper authorship space is only useful against forgeries if training also supplies examples of what a forgery is; conversely, hard negatives are only fully exploitable when the style space is fine-grained enough to place imitations away from a well-modeled centroid. Identity diversity’s *standalone* payoff is content-invariance (§9.4), an orthogonal axis. The production model needs both levers because it faces both failure modes.

9.2 Training-seed reproduction

The headline could be a lucky initialization. Retraining v14b from scratch with a different seed (new weight init and data order) reproduces the ranking within probe noise: AUC 0.976 ± 0.005 vs. 0.980 ± 0.003 , guardrail held for both (FPR_{other} 0.038 ± 0.017 vs. 0.079 ± 0.014). One honest nuance:

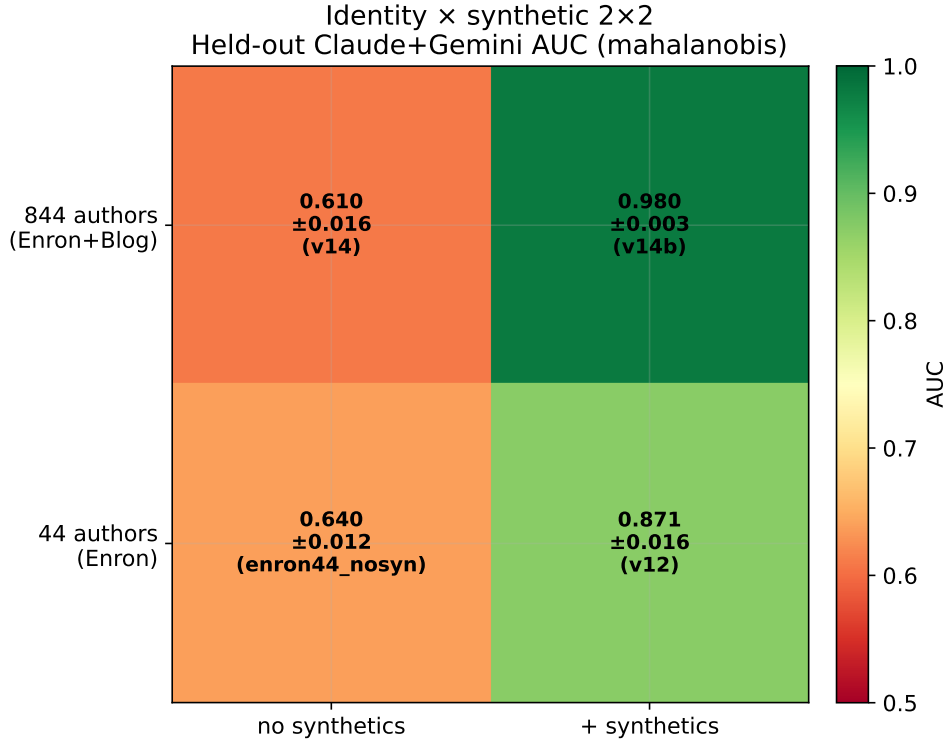


Figure 4: The completed 2x2. The interaction is super-additive: the whole (0.980 ± 0.003) exceeds the sum of the parts.

the operating-point tail carries more training-seed variance than AUC — TPR@5% moves from 0.909 ± 0.017 to 0.867 ± 0.037 across the two trainings. We therefore report TPR@5% as a band (~ 0.87 – 0.91) rather than a point, and recommend per-deployment threshold calibration.

9.3 Novel-vendor spot-check

Held-out Claude+Gemini could in principle have been “overfit” through repeated evaluation across the project. We generated 343 fresh forgeries with **Qwen-2.5-72B** and **DeepSeek-V3** — vendors outside both the training set and the recurring evaluation set — and evaluated blind:

Version	AUC (forgery/genuine)	TPR@5%	AUC g/other
v12	0.922 ± 0.005	0.630 ± 0.030	0.964 ± 0.000
v14b	0.996 ± 0.001	0.986 ± 0.008	0.966 ± 0.003

Table 6: Novel vendors (Qwen-2.5-72B + DeepSeek-V3), never in training or prior evaluation. Mahalanobis, mean±std over 3 probe seeds.

v14b’s separation is, if anything, *stronger* on the novel vendors — it did not overfit to Claude/Gemini. One caveat we report deliberately: on this pool the raw $\text{FPR}_{\text{other}}@5$ reads 0.58 ± 0.21 . This is a threshold-placement artifact, not a real leak — with synthetics near-perfectly separated (AUC 0.996), the 5%-synthetic-anchored threshold lands in a degenerate flat region of the score distribution (the ± 0.21 variance is the tell), while wrong-human *ranking*

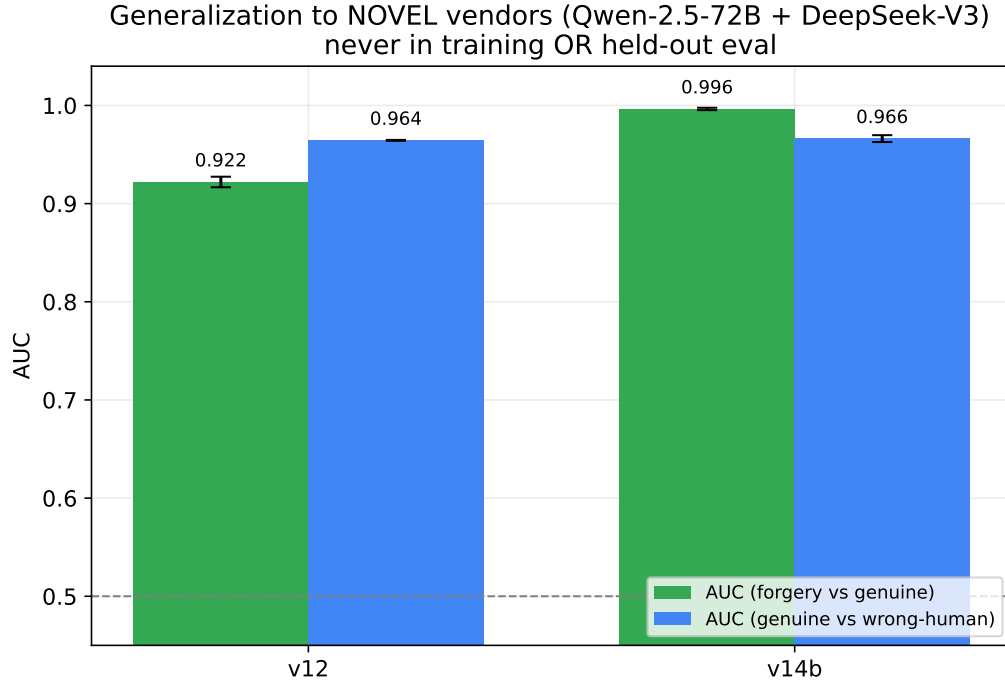


Figure 5: Novel-vendor spot-check (Qwen-2.5-72B + DeepSeek-V3): forgery separation (left) and genuine-vs-wrong-human ranking (right), v12 vs. v14b.

remains clean (AUC 0.97). This is the strongest demonstration in the paper of the §8.2 rule: as synthetic separation approaches saturation, a synthetic-anchored operating point becomes meaningless, and the threshold must be anchored on the wrong-human axis.

9.4 Content-invariance

Identity diversity’s independent payoff. On PAN cross-topic verification (clean out-of-domain — never in training for any version): v12/v13-recipe 0.779 → v14 0.879 → v14b 0.857. On the held-out 150-author blog probe: v12 0.786 → v14b 0.909 (corroborating, though blog is partly in-domain for v14/v14b since other blog authors are in training). v14b gives back a small part of v14’s PAN gain (0.879 → 0.857) in exchange for the forgery axis — an acceptable trade, and both far exceed the 0.779 plateau that held from v11 through v13.

10 Limitations

- **Sender ceiling.** The cross-generator probe has 44 profiled senders (the post-filter maximum of the Enron corpus). AUC and TPR@5% are tight, but TPR@1% retains wide intervals (± 0.05 – 0.06 even at the 5-seed mean); enlarging beyond 44 requires a second enrollment corpus. Lead with AUC and TPR@5%.
- **Length invariance is the weakest axis.** Verifying a short text against a long one remains the worst slice in every version. On enrollment-free pairwise probes over unseen senders (v11-era), mixed-length verification was near chance (0.53–0.64); on the current

enrolled probe v14b reaches AUC 0.930 on mixed-length pairs but only TPR@5% 0.71 — its weakest slice. The scoring-side mitigation we tested (length-matched enrollment) was cleanly refuted (it *discards* length-invariant signal; pAUC@5% -0.08). The open levers are length-conditioned thresholds and cross-length hard negatives in training.

- **Operating-point tail variance.** TPR@5% varies $\sim \pm 0.04$ across training seeds (§9.2); quote it as a band and calibrate thresholds per deployment.
- **Blog is partly in-domain.** The clean content-invariance evidence is PAN cross-topic (0.857); the blog number (0.909) corroborates but shares a corpus with training.
- **Adaptive attackers.** We evaluate few-shot imitation, not an attacker who iterates against the deployed detector’s scores. Style-space evasion under query access is untested.
- **Corpus and language scope.** Training and evaluation are English corporate email and blogs; transfer to other languages and registers is unmeasured.

11 Deployment recommendations

1. **Ship** the v14b checkpoint with the **Mahalanobis** scorer (cosine fallback and abstain below $k=5$ enrollment emails).
2. **Anchor the operating point on min(wrong-human, synthetic)** — never on the synthetic pool alone (§8.2, §9.3). Expected operating behavior at the 5% budget: forgery catch rate 0.87–0.91, wrong-human false-accept ≤ 0.10 .
3. **Calibrate per deployment.** The tail is threshold-sensitive; fit the operating point on the deployment’s own genuine/impostor score distributions. Per-sender or conformal calibration is the natural next step.
4. **Monitor for novel generators.** The novel-vendor result (AUC 0.996) is encouraging, but new vendors should be spot-checked as they appear — generation of a fresh imitation pool for a new vendor is an hours-scale, eval-only exercise in this codebase.
5. **Never retrain on deployment data silently.** The encoder is enrollment-only at deployment by design; profile updates are $O(1)$ per email and auditable.

12 Conclusion

The project’s thesis, earned the long way: **better authorship modeling is what catches LLM imitations**. Detectors of machine-generated text — trained or zero-shot — fail on few-shot imitation because the attacker’s text is engineered to be human-like; what the attacker cannot engineer, without being the victim, is the victim’s idiolect. Making that signal robust required two training levers with a super-additive interaction: synthetic hard negatives from diverse generators teach the model what forgery looks like, and identity diversity sharpens the style space until imitations have nowhere to hide — lifting held-out forgery detection from AUC 0.69 to 0.980 ± 0.003 while holding the wrong-human guardrail, transferring to never-seen vendors, and improving cross-topic authorship verification as a side effect.

Future work: length invariance (cross-length negatives, length-conditioned thresholds); a second enrollment corpus to break the 44-sender evaluation ceiling; per-sender/conformal threshold calibration; adaptive-attacker evaluation; and continuous novel-vendor monitoring.

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A Training configuration (v14b)

Backbone	LUAR-MUD (rrivera1849/LUAR-MUD), episode pooling, frozen
Adapter	LoRA $r=16$, $\alpha=32$, dropout 0.05, on q/k/v projections
Projection / max length	128-d, 512 tokens
Loss	episodic SupCon, $\tau=0.05$, support $k \in [2, 6]$, SupCon weight 0.5
Batch	$P \times K$ sampler, $K=8$, batch size 128, 4 synthetic pairs/batch
Augmentation	random crop $p=0.3$ (5–60 words)
Data	844 authors: 44 Enron senders + 800 Blog Authorship authors
Synthetics	few-shot imitations by GPT-4o-mini + Llama-3.1-70B
Optimization	lr 2×10^{-4} , cosine schedule, 200 warmup steps, grad clip 1.0, AMP, 80 epochs
Checkpoint monitor	$\min(\text{pAUC}_{\text{other}}, \text{pAUC}_{\text{syn}})$ at 5% FPR (maximize)
Seeding	--seed controls torch/numpy/python + samplers; seed 0 $\equiv$ historical runs

Table 7: v14b training configuration (`configs/experiments/v14b_manyauthor_syn_lora.yaml`).

B Reproduction

All tables regenerate from committed result JSONs; evaluation is minutes-scale, training is hours-scale on a single A40.

- Multi-seed held-out evaluations:
`bash scripts/whitepaper/run_multiseed_eval.sh → results/whitepaper/multiseed/`
- Novel-vendor pool + evaluation:
`bash scripts/whitepaper/gen_novelvendord.sh → results/whitepaper/novelvendord/`
- Trainings (v14b seed-1 reproduction; 2×2 missing corner):
`bash scripts/whitepaper/run_training.sh`
- Aggregation and figures:
`python scripts/whitepaper/aggregate.py; python scripts/whitepaper/make_figures.py;`
`python scripts/make_v14b_figures.py`
- Failed pivots:
`scripts/eval_zeroshot_detector.py, scripts/eval_style_embedding.py → results/v14/`

C Full multi-seed table (both scorers)

Scorer	Version	AUC	TPR@5%	TPR@1%	FPR _{other} @5	AUC g/other
linear_z3	v11	0.705 $\pm$ 0.021	0.135 $\pm$ 0.031	0.032 $\pm$ 0.018	0.000 $\pm$ 0.000	0.932 $\pm$ 0.004
	v12	0.872 $\pm$ 0.015	0.462 $\pm$ 0.078	0.150 $\pm$ 0.018	0.004 $\pm$ 0.002	0.931 $\pm$ 0.011
	v13	0.875 $\pm$ 0.016	0.523 $\pm$ 0.085	0.256 $\pm$ 0.161	0.007 $\pm$ 0.002	0.940 $\pm$ 0.006
	v14	0.599 $\pm$ 0.018	0.088 $\pm$ 0.014	0.033 $\pm$ 0.018	0.023 $\pm$ 0.013	0.647 $\pm$ 0.016
	v14b	0.965 $\pm$ 0.005	0.943 $\pm$ 0.013	0.555 $\pm$ 0.078	0.253 $\pm$ 0.032	0.940 $\pm$ 0.008
	v14b (train seed 1)	0.962 $\pm$ 0.004	0.919 $\pm$ 0.024	0.436 $\pm$ 0.078	0.203 $\pm$ 0.043	0.939 $\pm$ 0.006
	44-author, -syn	0.658 $\pm$ 0.016	0.081 $\pm$ 0.018	0.022 $\pm$ 0.010	0.000 $\pm$ 0.000	0.943 $\pm$ 0.003
mahalanobis	v11	0.690 $\pm$ 0.020	0.121 $\pm$ 0.017	0.043 $\pm$ 0.012	0.000 $\pm$ 0.000	0.967 $\pm$ 0.003
	v12	0.871 $\pm$ 0.016	0.453 $\pm$ 0.064	0.155 $\pm$ 0.036	0.001 $\pm$ 0.001	0.965 $\pm$ 0.002
	v13	0.872 $\pm$ 0.021	0.499 $\pm$ 0.069	0.255 $\pm$ 0.047	0.001 $\pm$ 0.001	0.970 $\pm$ 0.004
	v14	0.610 $\pm$ 0.016	0.137 $\pm$ 0.015	0.060 $\pm$ 0.013	0.010 $\pm$ 0.003	0.667 $\pm$ 0.015
	v14b	0.980 $\pm$ 0.003	0.909 $\pm$ 0.017	0.533 $\pm$ 0.055	0.079 $\pm$ 0.014	0.967 $\pm$ 0.003
	v14b (train seed 1)	0.976 $\pm$ 0.005	0.867 $\pm$ 0.037	0.493 $\pm$ 0.052	0.038 $\pm$ 0.017	0.967 $\pm$ 0.007
	44-author, -syn	0.640 $\pm$ 0.012	0.134 $\pm$ 0.025	0.045 $\pm$ 0.019	0.000 $\pm$ 0.000	0.982 $\pm$ 0.002

Table 8: Complete multi-seed results (5 probe seeds), both scorers, all seven trained models (regenerated by `scripts/whitepaper/aggregate.py`).